

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I Donthu Raghava-192525304 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Donthu Raghava

Assessment Tasks - Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients.

Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

1. Understanding of the Case

The retail chain requires a cloud-based inventory management system that connects multiple stores through centralized cloud services. The system should:

- Manage inventory across all stores.
- Connect POS systems and management dashboards.
- Store product information and transaction records securely.
- Perform demand forecasting using cloud analytics.
- Provide secure communication using VPN, Firewall, and API Gateway.
- Offer high availability and scalability.

2. Application of Cloud Concepts

Cloud Service Models

Service	Purpose
IaaS	Virtual machines for backend inventory services
PaaS	Analytics platform and demand forecasting
SaaS	Web dashboard for store managers

Cloud Components

- POS terminals (Client)
- Management Dashboard
- REST APIs
- API Gateway
- Cloud Database

- Cloud Storage
- Analytics Platform
- Demand Forecasting Service
- VPN
- Firewall
- Load Balancer

3. Analysis

Advantages

- Centralized inventory across all branches
- Real-time stock updates
- Accurate demand prediction
- Reduced manual inventory errors
- Easy expansion to new stores

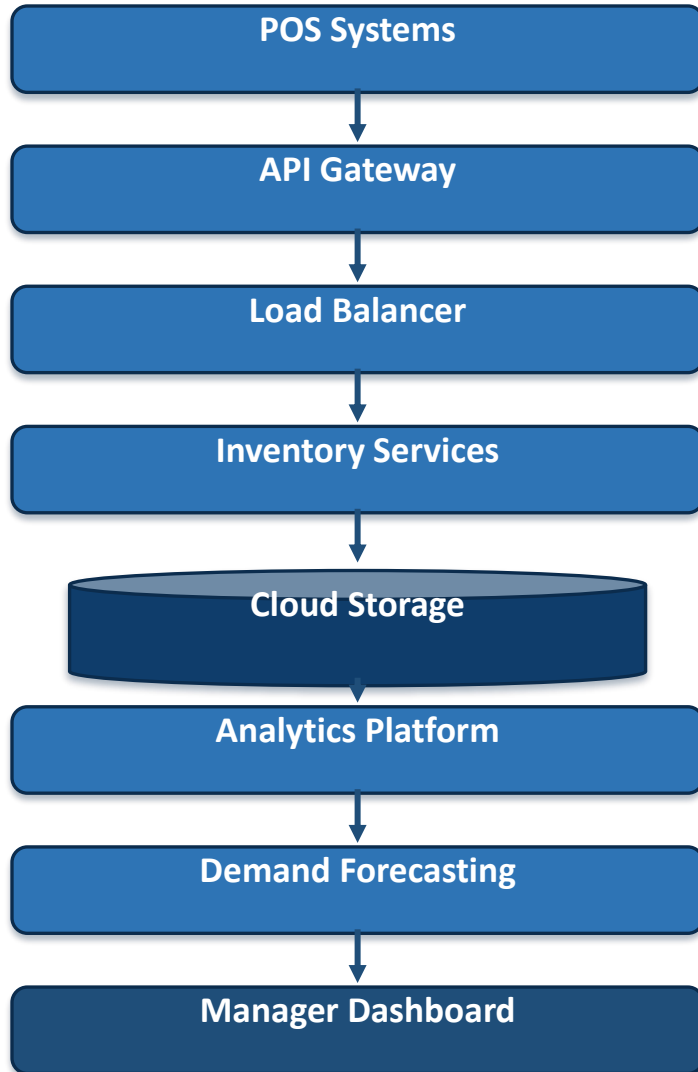
Challenges

- Internet dependency
- Cloud service cost
- Data security
- API latency

Trade-offs

Choice	Benefit	Limitation
Centralized Database	Consistent inventory	Higher network dependency
Cloud Analytics	Better forecasting	Additional processing cost
API Gateway	Improved security	Slight latency

4. Solution Design



Security

- VPN
- Firewall
- HTTPS
- API Authentication
- Role-Based Access Control (RBAC)

5. Conclusion

The proposed architecture provides centralized inventory management, real-time updates, scalable cloud infrastructure, and secure communication while enabling analytics-driven business decisions.

Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.

Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

1. Understanding of the Case

The enterprise requires an online document collaboration platform that enables:

- Real-time editing
- Document sharing
- Version control
- Synchronization across devices
- Global availability
- Secure collaboration

2. Application of Cloud Concepts

Cloud Service Models

Service	Purpose
SaaS	Document editing application
PaaS	Collaboration engine
IaaS	Compute instances for backend processing

Components

- Web Browser
- Mobile Client
- REST/WebSocket APIs
- Collaboration Server
- Distributed Cloud Storage
- Version Control
- Identity Management
- Load Balancer

3. Analysis

Advantages

- Simultaneous editing
- Automatic saving
- Version history
- Access from anywhere
- High availability

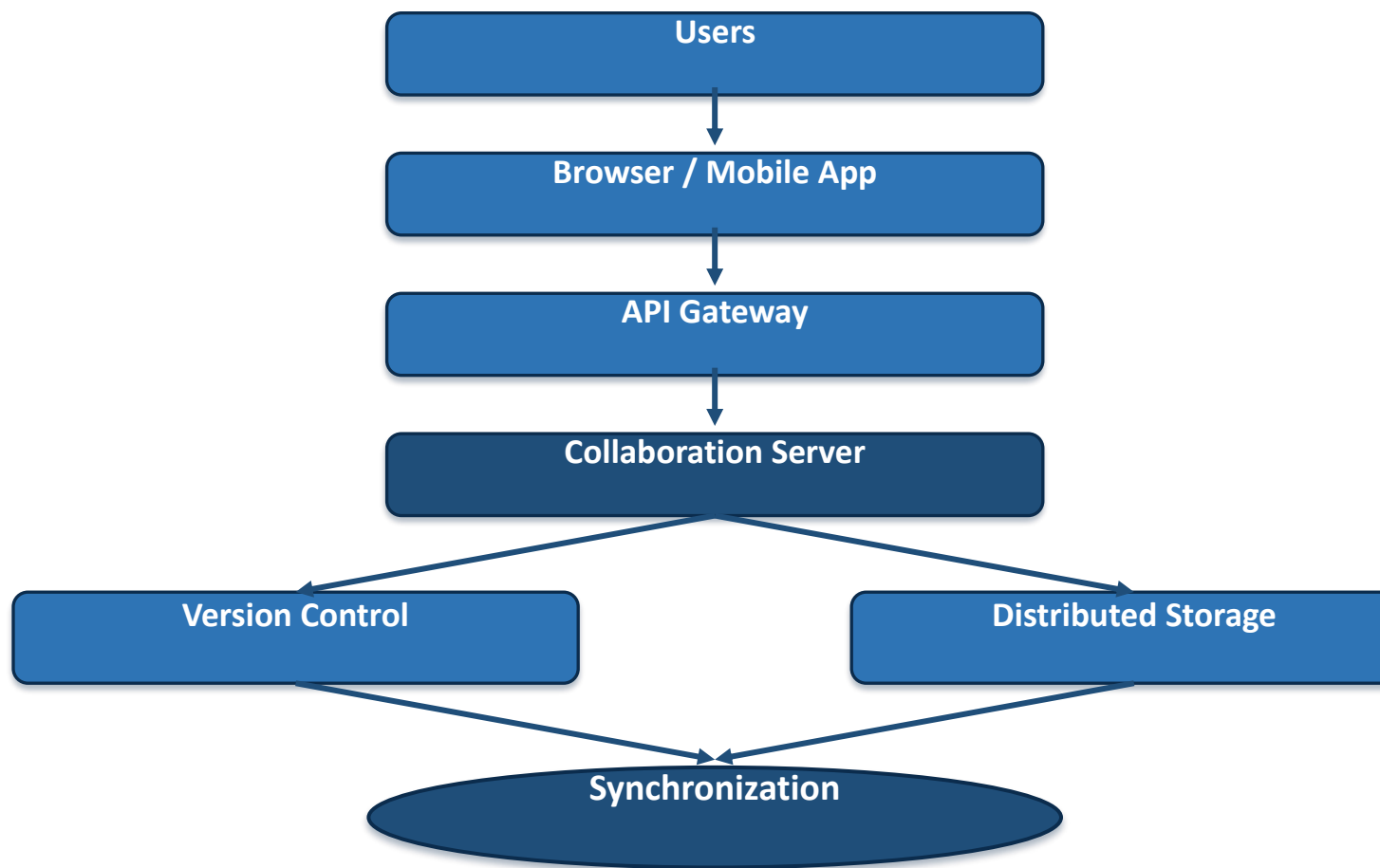
Challenges

- Conflict resolution
- Data synchronization
- Large storage requirements
- Security threats

Trade-offs

Choice	Benefit	Limitation
Distributed Storage	High availability	More complex management
Real-time Sync	Better collaboration	Increased network usage
Encryption	Better security	Small performance overhead

4. Solution Design



Security

- HTTPS
- Identity Management
- Access Permissions
- Encryption at Rest
- Encryption in Transit
- Multi-Factor Authentication

5. Conclusion

The cloud-based collaboration system enables secure, scalable, and highly available document editing with efficient synchronization and reliable version control.

Case Study 3: Online Food Ordering Platform

1. Understanding of the Case

The startup requires an online food ordering platform supporting:

- Mobile and web clients
- RESTful APIs
- Cloud storage
- Scalable backend
- Secure payments
- Reliable infrastructure

2. Application of Cloud Concepts

Cloud Service Models

Service Purpose

IaaS Virtual Machines

PaaS Managed Kubernetes

SaaS Food ordering web application

Components

- Mobile App
- Web Browser
- REST APIs
- API Gateway
- Kubernetes Cluster
- Cloud Storage

- Database
- Authentication Server
- Payment Service

3. Analysis

Advantages

- Supports thousands of users
- Easy deployment
- High scalability
- Fault tolerance
- Secure transactions

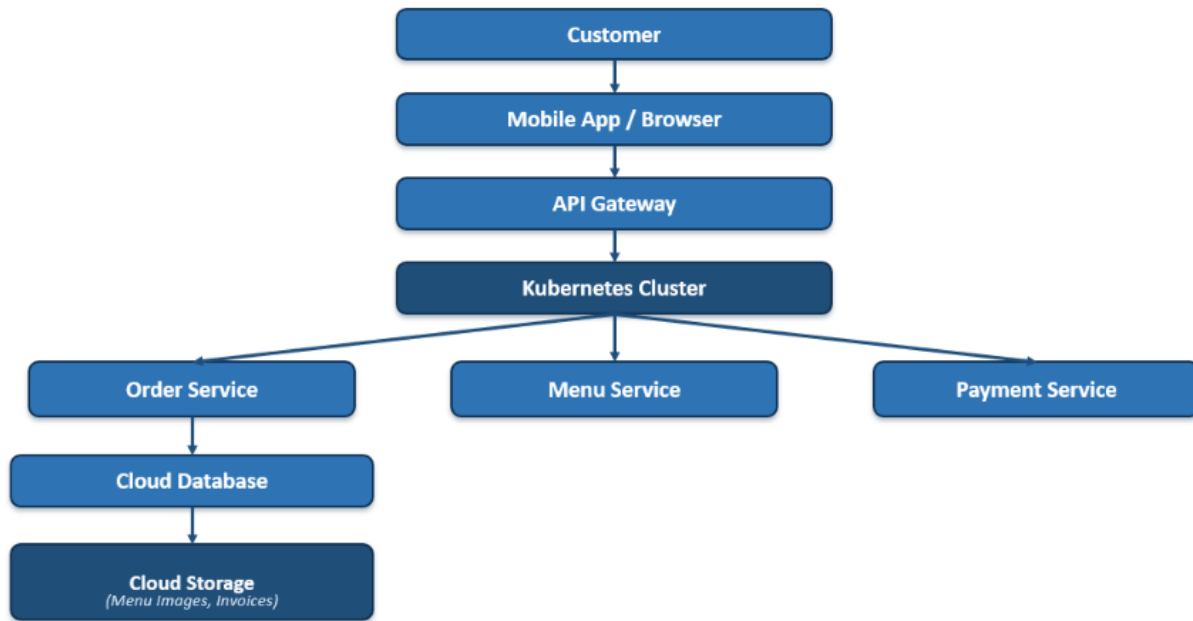
Challenges

- Traffic spikes
- Payment security
- Storage management
- Service monitoring

Trade-offs

Choice	Benefit	Limitation
Kubernetes	Automatic scaling	Higher management complexity
Cloud Storage	Secure file storage	Storage cost
REST APIs	Easy integration	Stateless communication

4. Solution Design



Security

- HTTPS
- JWT Authentication Tokens
- Role-Based Access Control (RBAC)
- OAuth Authentication
- Firewall
- API Gateway

5. Conclusion

The proposed architecture provides a scalable, secure, and highly available food ordering platform capable of handling real-time customer requests while maintaining data security and efficient cloud resource utilization.